

d.RADIUS & TACACS+ : TACACS+ & RADIUS Features, RADIUS Authentication Process, TACACS+ Use Cases, RADIUS vs. TACACS+

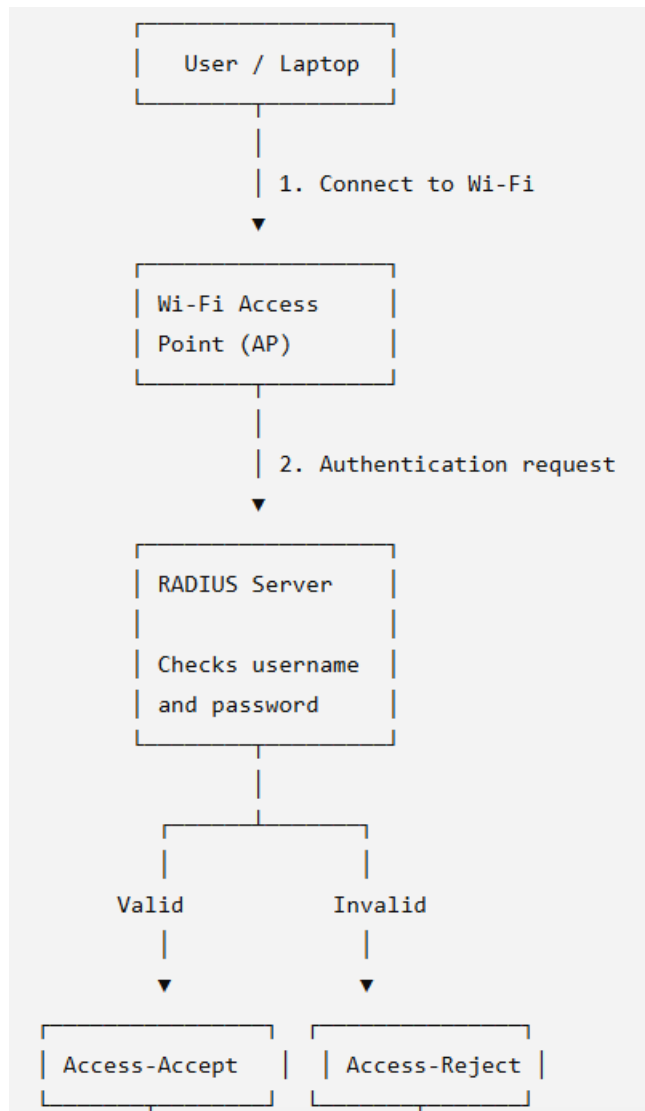
Task 1.

Create a comparison table listing at least 4 key differences between RADIUS and TACACS+ features, focusing on protocol type, encryption, ports used, and command authorization.

Feature	RADIUS	TACACS+
Protocol type	UDP-based protocol	TCP-based protocol
Encryption	Encrypts the user's password; other packet information is generally not encrypted	Encrypts the entire TACACS+ packet payload
Ports used	UDP 1812 for authentication and 1813 for accounting	TCP 49
Command authorization	Limited command-level authorization; commonly used for network access authentication	Provides detailed per-command authorization
AAA separation	Authentication and authorization are commonly combined	Authentication, authorization, and accounting are separated
Typical use	Wi-Fi authentication, VPN, network access	Administration of routers, switches, firewalls and other network devices

Task 2.

Draw a simple flowchart or sequence diagram that illustrates the RADIUS authentication process for a user trying to log in to a Wi-Fi network (like at a coffee shop or college campus).



Sequence

1. The user connects to the college/coffee-shop Wi-Fi.
2. The Access Point (AP) acts as the network access device.
3. The AP sends the user's authentication request to the RADIUS server.
4. The RADIUS server verifies the user's credentials.
5. If valid, the server sends Access-Accept.

6. The AP allows the user onto the network.
7. If credentials are invalid, the server sends Access-Reject and access is denied.

Task 3.

Describe a real-world scenario from any app you use (such as Zomato, Paytm, or IRCTC) where TACACS+ would be more suitable than RADIUS for managing admin access, and explain why.

- Consider an organization such as Zomato with network administrators managing routers, switches, firewalls and other infrastructure.
- Suppose an administrator logs in to a Cisco router.
- TACACS+ can authenticate the administrator and then determine which commands that administrator is allowed to execute.

An administrator might be permitted to:

show ip interface brief

show running-config

but another account might not be permitted to execute configuration-changing commands such as:

configure terminal

Why TACACS+ is suitable

TACACS+ is useful for administrative access because it provides granular command authorization. It can help an organization control what different administrators can do on network devices and maintain accounting records of administrative activity.

RADIUS is more commonly associated with network-access authentication, such as Wi-Fi, VPN, and 802.1X.

Task 4.

List two advantages and two disadvantages of using RADIUS for network authentication in a public hotspot (like those found in malls or airports).

Advantages

1. Centralized authentication

A mall, airport, or college can use one RADIUS server to authenticate users across multiple access points.

This makes user management easier than maintaining separate credentials on every AP.

2. Supports large numbers of users

RADIUS is designed for network-access authentication and can be integrated with Wi-Fi/802.1X infrastructure, making it suitable for environments with many users and access points.

Disadvantages

1. RADIUS server dependency

If the RADIUS server or its connectivity becomes unavailable, users may be unable to authenticate.

2. Configuration and security management

Deploying RADIUS requires proper configuration of the access points, server, shared secrets, certificates where applicable, and user authentication infrastructure. Poor configuration can create security weaknesses.

Task 5.

Use ChatGPT or another AI tool to generate a sample TACACS+ configuration for a Cisco router, then review the output and highlight any commands you would modify for a college lab setup.

Hint: Ask the AI tool for 'basic TACACS+ configuration for Cisco router' and check for lines like 'tacacs-server host' and 'aaa new-model'.

```
enable
configure terminal

! Enable AAA
aaa new-model

! Configure TACACS+ server
tacacs server TACACS-SERVER
  address ipv4 192.168.10.10
  key cisco123

! Create AAA authentication method
aaa authentication login default group tacacs+ local

! Configure authorization
aaa authorization exec default group tacacs+ local

! Configure VTY lines
line vty 0 4
  login authentication default
  transport input telnet ssh

end
write memory
```

Command	Sample value	What I would change
address ipv4 192.168.10.10	192.168.10.10	Change it to the actual IP address of the TACACS+ server in the Packet Tracer/lab topology
key cisco123	cisco123	Use the same shared secret configured on the TACACS+ server
aaa authentication login default group tacacs+ local	TACACS+ + local fallback	Keep this because local provides a fallback if the TACACS+ server is unavailable
transport input telnet ssh	Telnet + SSH	Prefer transport input ssh in a security-focused lab
line vty 0 4	VTY 0–4	Keep if the lab uses these VTY lines; the exact range depends on the router model